



SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR

Constituent of Symbiosis International (Deemed University), Pune

(Established under Section 3 of the UGC Act of 1956 wide notification number F-9-12/2001-U-3 of Government of India)

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Computer Networks/0705210503

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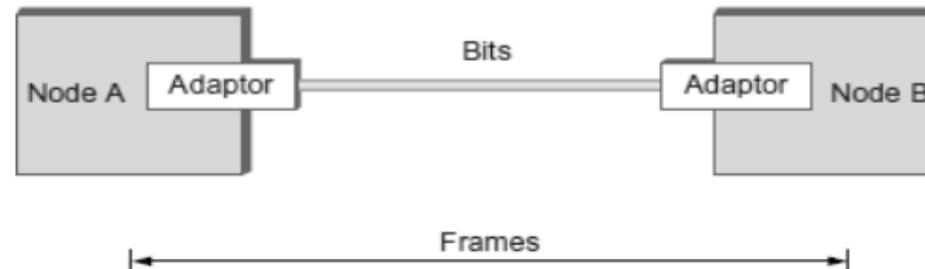
3. DLC SERVICES

- The data link control (DLC) deals with procedures for communication between two adjacent nodes—node-to-node communication—no matter whether the link is dedicated or broadcast.
- Data link control service include
(1) Framing (2) Flow Control (3) Error Control

1.

FRAMING

- The data-link layer packs the bits of a message into frames, so that each frame is distinguishable from another.



- Although the whole message could be packed in one frame, that is not normally done.
- One reason is that a frame can be very large, making flow and error control very inefficient.



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- When a message is carried in one very large frame, even a single-bit error would require the retransmission of the whole frame.
- When a message is divided into smaller frames, a single-bit error affects only that small frame.
- Framing in the data-link layer separates a message from one source to a destination by adding a sender address and a destination address.
- The destination address defines where the packet is to go; the sender address helps the recipient acknowledge the receipt.

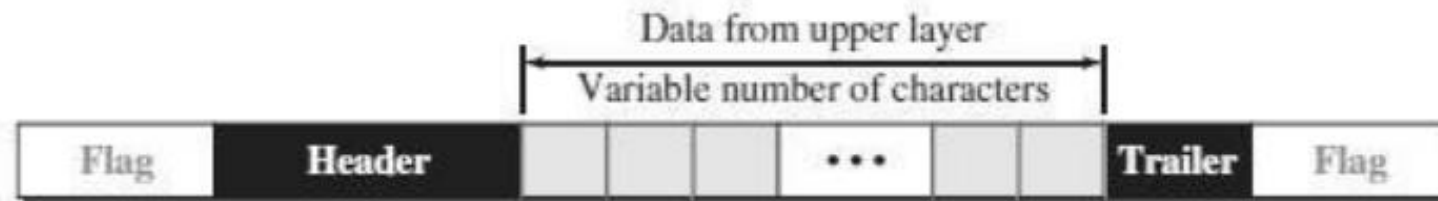
Frame Size

- Frames can be of fixed or variable size.
- Frames of fixed size are called cells. In fixed-size framing, there is no need for defining the boundaries of the frames; the size itself can be used as a delimiter.
- In variable-size framing, we need a way to define the end of one frame and the beginning of the next. Two approaches were used for this purpose: a character-oriented approach and a bit-oriented approach.



Character-Oriented Framing

- In character-oriented (or byte-oriented) framing, data to be carried are 8-bit characters.
- To separate one frame from the next, an 8-bit (1-byte) flag is added at the beginning and the end of a frame.
- The flag, composed of protocol-dependent special characters, signals the start or end of a frame.



- Any character used for the flag could also be part of the information.
- If this happens, when it encounters this pattern in the middle of the data, the receiver thinks it has reached the end of the frame.
- To fix this problem, a **byte-stuffing** strategy was added to character-oriented framing.



Byte Stuffing (or) Character Stuffing

- Byte stuffing is the process of adding one extra byte whenever there is a flag or escape character in the text.
- In byte stuffing, a special byte is added to the data section of the frame when there is a character with the same pattern as the flag.
- The data section is stuffed with an extra byte. This byte is usually called the escape character (ESC) and has a predefined bit pattern.
- Whenever the receiver encounters the ESC character, it removes it from the data section and treats the next character as data, not as a delimiting flag.

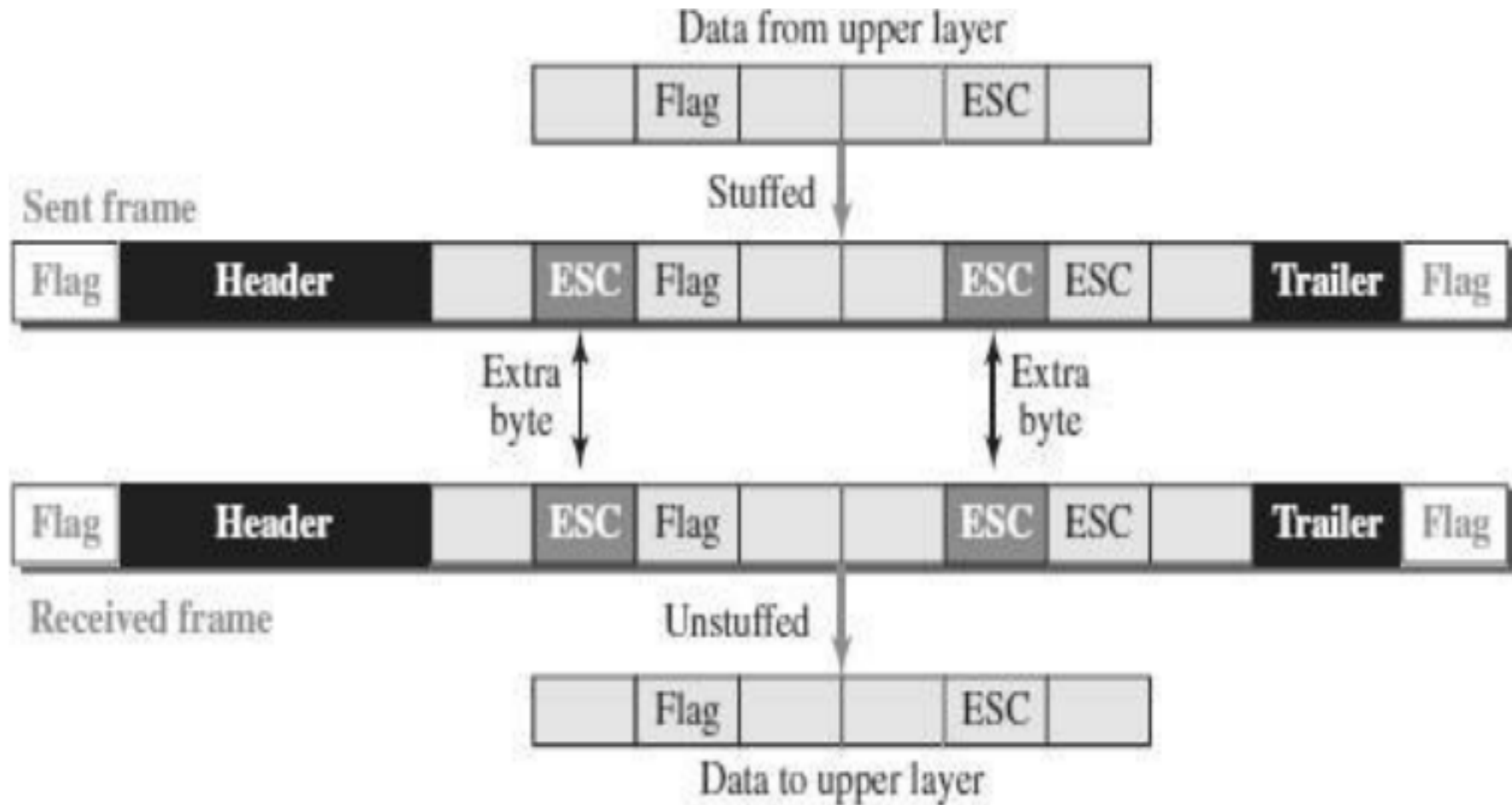


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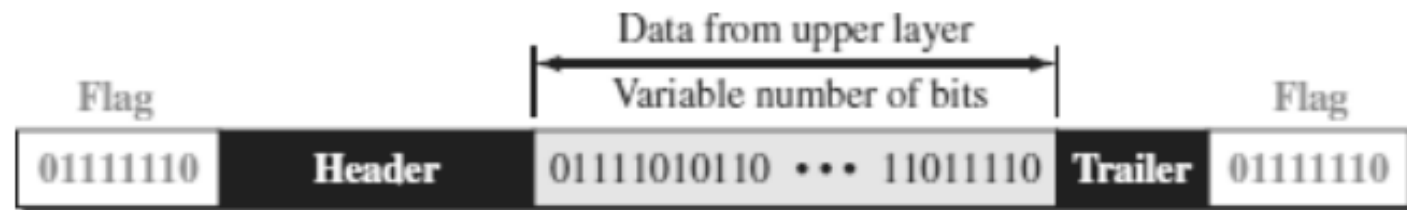
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Bit-Oriented Framing

- In bit-oriented framing, the data section of a frame is a sequence of bits to be interpreted by the upper layer as text, graphic, audio, video, and so on.
- In addition to headers and trailers), we still need a delimiter to separate one frame from the other.
- Most protocols use a special 8-bit pattern flag, 01111110, as the delimiter to define the beginning and the end of the frame



- If the flag pattern appears in the data, the receiver must be informed that this is not the end of the frame.
- This is done by stuffing 1 single bit (instead of 1 byte) to prevent the pattern from looking like a flag. The strategy is called **bit stuffing**.



Bit Stuffing

- Bit stuffing is the process of adding one extra 0 whenever five consecutive 1s follow a 0 in the data, so that the receiver does not mistake the pattern 0111110 for a flag.
- In bit stuffing, if a 0 and five consecutive 1 bits are encountered, an extra 0 is added.
- This extra stuffed bit is eventually removed from the data by the receiver.
- The extra bit is added after one 0 followed by five 1's regardless of the value of the next bit.
- This guarantees that the flag field sequence does not inadvertently appear in the frame.



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